

MID SEM - I (PHIC 101, R&IA, CE, EE, SET, ECE, ME, PIE)

TIME: 4:00 PM - 4:50 PM

B.Tech (1st Semester)

Time Allowed: 50 min

Max. Marks: 15

NOTE: Attempt any **Three** questions. Extra ANSWER SHEETS will **NOT** be supplied.

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- (1) Discuss following processes in LASERS with neat relevant diagrams, (i) spontaneous emission (ii) population inversion. Discuss briefly that population inversion is unachievable in a two level system by increasing the photon flux. [2+2+1]
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- (2) Define the coherence length? Calculate the coherence length of LASER beams having band width 500 Hz and 1.0 GHz. Which one is suitable for producing diffraction pattern? [2+2+1]
- (3) Tabulate the differences between single mode and multimode optical fiber with relevant diagrams. What happens to the light propagation when optical fiber is bended by an angle 90° ? [4+1]
- (4) Write inverse Lorentz transformation equations. Derive the expression for Time dilation. What is twin paradox, state in brief? [1+3+1]
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